

Chapter 5

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet for Chapter 5: Critical Thinking and Problem-Solving in the AI Era

This worksheet is designed for high school students (Class 9-12) as part of the course "AI & Future Work Preparedness." It includes a variety of activities to reinforce key concepts from Chapter 5, encourage critical thinking, and engage students in practical applications of the material. The worksheet can be used for individual or group work, in class or as homework.

Part 1: Concept Mapping (Visual Organizer)

Create a small concept map or diagram to connect key ideas from the chapter. In the center, write "Critical Thinking in AI Era." Draw at least 3 branches coming out of the center, each labelled with a key concept (e.g., Bias in AI, Creative Problem-Solving, Ethical Decision-Making). For each branch, write 1-2 short phrases or keywords explaining its importance or relevance. Use the space below for your map or attach a separate sheet if needed.

Space for Concept Map or Notes:

(If drawing, describe briefly what you've included if not attaching a separate sheet.)

Part 2: Practical Application (Scenario Analysis)

Read the scenario below and answer the questions to apply critical thinking and problem-solving skills.

Scenario: You and your friends are creating a school project using an AI tool to generate images for a presentation on climate change. However, you notice that the AI keeps generating images that mostly show impacts in certain regions while ignoring others, potentially creating a biased view of the issue.

- **Identify the Problem:** What issue do you see with the AI tool in this scenario?
- **Critical Thinking:** What questions would you ask to understand why the AI is producing these results?
- **Creative Solution:** Suggest one way to solve or work around this bias using technology or another method.
- **Decision-Making:** If you had to decide whether to use these images or find alternatives, what factors would you consider?

Part 3: Fun Activity (Group or Individual)

"AI Bias Detective" Challenge Pick an app or platform you use daily (like YouTube, Instagram, or a music app). For one day, observe the recommendations or content it suggests to you. Answer the following:

- **App Chosen:** _____
- **What patterns did you notice in the recommendations? (e.g., same genre, same type of content)**
- **Do you think there's bias in what's being shown? Why or why not?**
- **How can you "break the bias bubble"? Suggest one action to see different content.**

Optional Bonus: Share your findings with a classmate or group and compare results. Did you notice similar biases? Write a quick note on what you learned from this comparison.

Part 4: Reflection and Future Thinking

Write a short paragraph (4-5 sentences) answering the following prompt: **How can I use critical thinking or problem-solving skills from this chapter in my daily life or future career?** Consider a specific situation, like evaluating online information, making decisions, or addressing tech bias.

Part 5: Creative Project (Optional Extension)

“Invent a Solution” Mini-Project Imagine a common teen problem (e.g., stress from exams, cyberbullying, or organizing group projects). Design a fictional AI tool or app to solve it using ideas from this chapter. Answer the following:

- **Problem:** What issue are you solving?
- **Solution Name:** What’s your AI tool/app called?
- **How It Works:** Describe how it uses technology to help.
- **Avoiding Bias:** How will you ensure it’s fair and inclusive for all users?